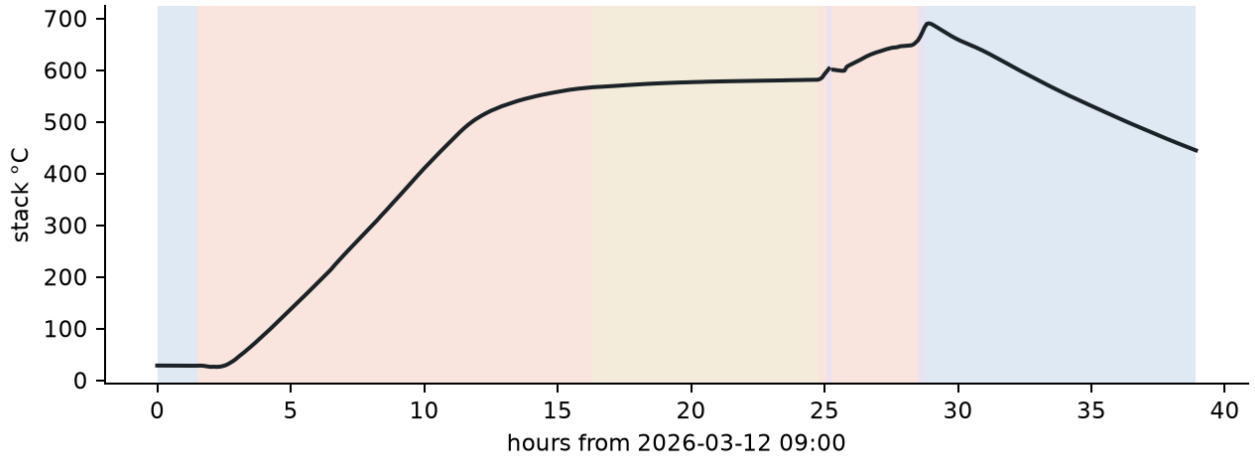
How the fuel-cell plant starts, stops and runs under load

Plain-language summary of the start-up, shut-down and part-load analysis notebooks · 2026-10-14

Overview

Three new analysis notebooks each take one part of the plant's day (warming up, cooling down, producing power) and turn all 39 logged days into numbers a data generator can copy.

Why it matters: the synthetic-data generator has to imitate these three behaviours. It now has measured recipes for each, plus a short list of decisions only you can make.



Real stack temperature over 39 hours. Bands: blue = shut-down, orange = start-up, gold = hot standby, purple = producing power.

Part	Share of time	Episodes	Headline
Start-up	177 h (19 %)	31	Heats at about 50 °C per hour; 12–15 h from cold
Shut-down	404 h (44 %)	29	Cooling speed is set by the air flow; 7 of 21 active ones began with a safety trip
Part load	4.4 h (0.5 %)	10	Short runs, capped by the weakest cell; 8 of 10 ended in a safety trip

Decisions needed from you

1. **Full (rated) power**: the data reach only 54 % of rated current. Leave rated power out of the generator, or take it from design values?

2. **False start on 2026-02-06**: 10.5 h labelled as heating although nothing ran. Raise the heater-on threshold from 0.2 kW to about 1 kW?

3. **Warm park at ~280 °C (03-20 → 03-22)**: give it one label of its own (for example "Warm hold")?

4. **Three bad logger minutes**: blank them in the labelling code too. They caused the false "207 °C/h" reading on 03-13.

5. **Plant document fix**: `operational-modes.md` lists 18.8 A on 03-20, but no load ran that day.

6. **Questions for the operator**: why the faster March cooling, the normal air-off temperature, the cause of the 03-25 08:50 emergency stop, and which exhaust limit (200 or 300 °C) to use.

Files: data_analysis/startup_analysis.ipynb → startup/ · data_analysis/shutdown_analysis.ipynb → shutdown/ · data_analysis/part_load_analysis.ipynb → part_load/ · shared loader: data_analysis/stage_signals.py

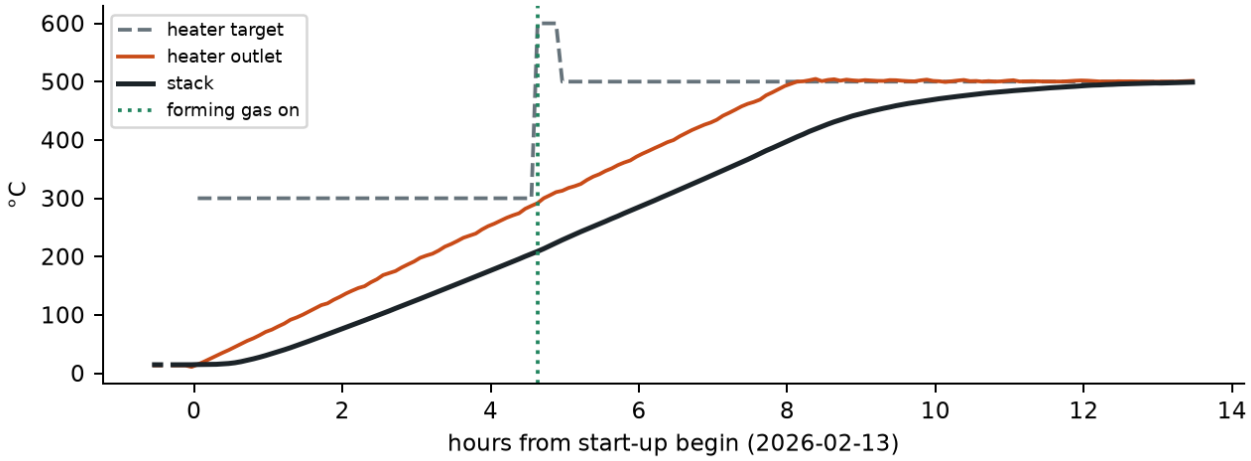
If you remember one thing: Each of the three plant behaviours now has its own measured recipe. The biggest open choice is what to do about full power, which the plant never reached.

Start-up: warming the stack

The plant warms the stack with an electric air heater on a fixed recipe, and the stack follows at about 50 °C per hour.

Steps: (1) air on, (2) heater outlet target rises 60 K per hour to 300 °C, (3) forming gas (nitrogen + a little hydrogen) starts when the outlet is at about 290 °C, (4) target raised to 500 or 570 °C, (5) hot standby after 12–15 h.

Odd findings: the 2026-02-06 'start-up' is a false start (nothing ran); the earlier '207 °C/h' on 03-13 came from bad logger minutes; the 274 °C hold on 03-22 was most likely a park at ~280 °C since 03-20.



A real cold start-up (2026-02-13): heater outlet follows its target, the stack trails by 55–75 K.

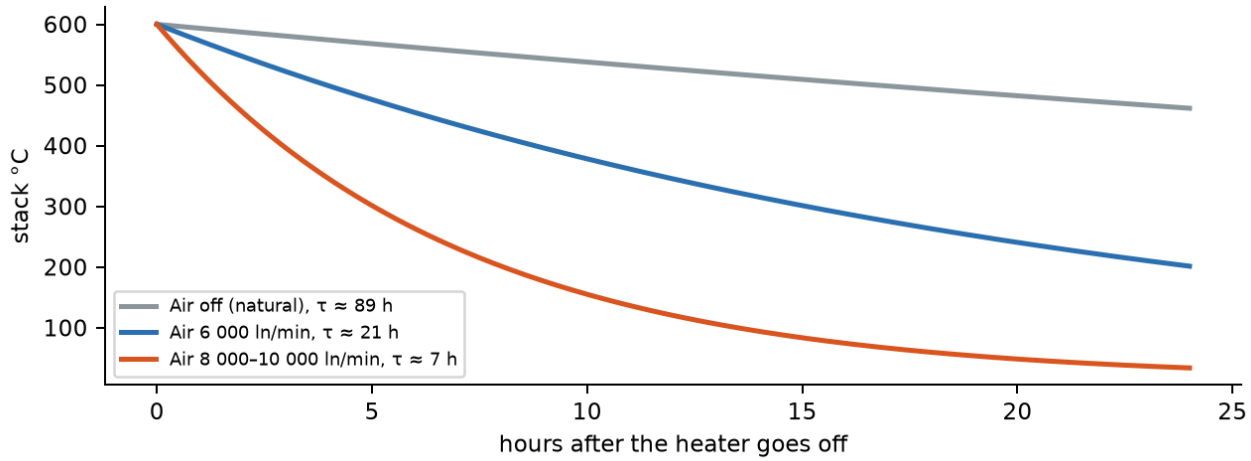
What	Value
Heating speed of the stack	≈ 50 °C per hour (100–500 °C)
Heater outlet ramp	60 K per hour (120 in some runs)
Stack lags the heater by	55–75 K
Forming gas switched on at	stack ≈ 210 °C (heater outlet ≈ 290 °C)
Heater energy for a full cold start	≈ 205 kWh to 495 °C
Fastest single thermocouple	above 120 °C/h only on 2025-11-13 (18 min)
Cold start-ups that reached the hold	4 of 8, in 12.3–14.7 h

If you remember one thing: A start-up is a slow, controlled ramp: heater outlet +60 K per hour, the stack about 50 K per hour behind it, forming gas at about 210 °C.

Shut-down: cooling the stack

The heater switches off and the air keeps blowing. How much air blows decides how fast the stack cools.

Order: power stops → heater off (after a trip: gas off, forming gas back within 2–7 min) → air cools the stack → forming gas off at 210–290 °C → air off at 200–260 °C, or kept on to near room temperature.



Cooling from 600 °C with the measured time constants.

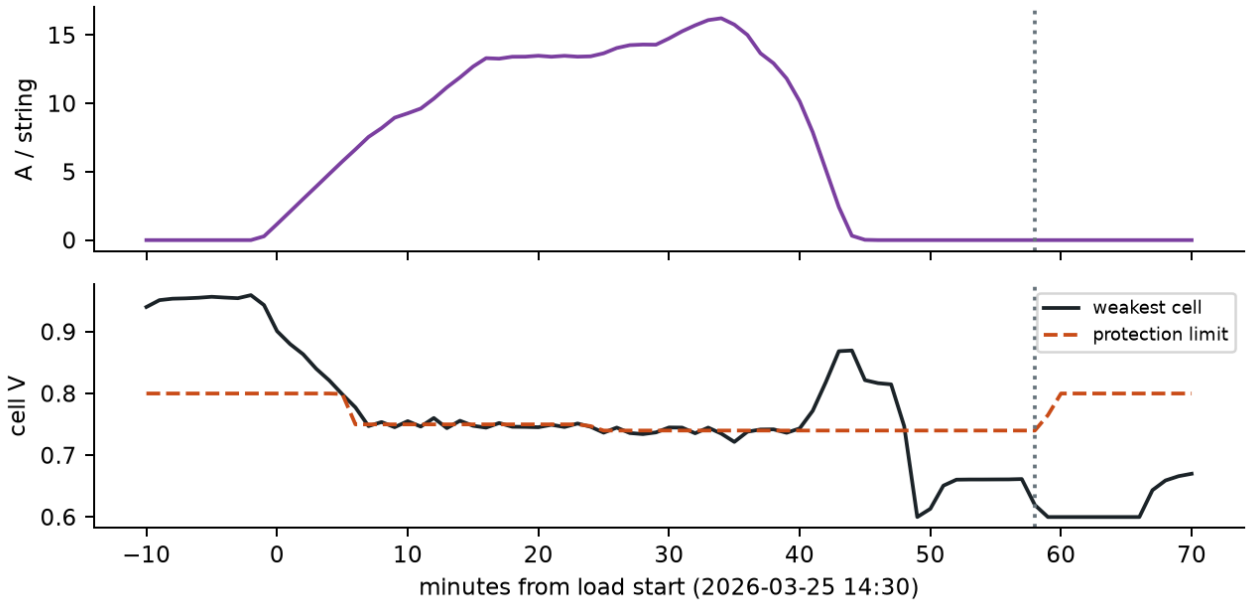
What	Value
Cooling speed, normal air (6 000 ln/min)	≈ –25 °C per hour at 600–400 °C
Cooling speed, strong air (8 000–10 000)	≈ –50 to –70 °C per hour
Cooling time constant τ	21 h / 5–9 h / 89 h with air off
Forming gas switched off at	210–290 °C
Air switched off at	200–260 °C, or kept on to 25–46 °C
Started by a safety trip	7 of 21 active shut-downs
Short cool-backs of a hot stack	10, dropping 6–94 K in 0.5–3.8 h

If you remember one thing: Cooling speed is set by the air: about –25 °C per hour with normal air and two to three times faster with the strong March setting.

Part load: producing power

The plant produced power only in ten short runs, and each time the current stopped rising because the weakest cell reached its safety limit.

Runs end early because the exhaust after the burner heats 2–4 °C per minute and reaches its 200 °C trip value (raised to 300 °C from 03-27).



A real run (2026-03-25 14:30): current rises in steps while the weakest cell sits on its protection limit.

What	Value
Runs	10, lasting 8–44 min, all in March 2026
Highest current / power	16.2 A per string / 33.6 kW (54–56 % of rated)
Current increase	0.42 A/min typical, never above 0.92 (limit 1)
What stops the current	weakest cell hits its limit (0.72–0.90 V)
Stack warming under load	+7 to +88 K per run
Exhaust temperature	+2–4 °C per minute, trip at 200 °C
Ended by a safety trip	8 of 10
Estimate at full current (30 A)	≈ 51 kW, weakest cell ≈ 0.58 V

If you remember one thing: Full power was never reached. The weakest cell caps the current near 15 A, and the exhaust temperature trips most runs within half an hour.